

From a ZX Spectrum in Bielawa to vibe coding: research for Pete Michaliszyn's technology chapter

Pete Michaliszyn's technology story traces a remarkable arc — from learning BASIC on a smuggled ZX Spectrum in a small Polish textile town where his grandfather worked at a legendary radio factory, through the wild bazaar-driven computing culture of 1990s Poland, to building his first app in 2026 using AI tools without writing a line of code himself. This research compiles the historical, cultural, and technical context needed to tell that story.

Diora: the radio factory 7 kilometers up the road

Diora was Poland's first post-war radio manufacturer, founded on **November 8, 1945** in **Dzierżoniów** [Oldradio](#) — a small city just **7 km north of Bielawa**. The name is an anagram of the Polish word "radio," [Naszemiasto](#) adopted after a factory-wide naming competition in 1957. [Oldradio](#) The factory was established by engineer Wilhelm Rotkiewicz [Letheko](#) in the abandoned buildings of a former German textile works that had been repurposed during the war by the German firm Hagenuk Werke Kiel to produce naval navigation equipment and radio stations. [Oldradio](#)

Diora grew into one of Poland's most significant electronics enterprises. At its peak, the factory **employed over 6,600 workers**, [Wedrownemisie](#) and during the 1960s, **two-thirds of all radio receivers in Poland** came from Dzierżoniów. [Naszemiasto](#) Its product line spanned decades of consumer electronics innovation: from the "Ludowy" receiver assembled from leftover German parts in 1945 (whose Bakelite casing had a Nazi eagle that had to be scraped off during production), [Oldradio](#) to the **Pionier** — the first entirely Polish-designed radio, of which over **one million units** were produced [Wikipedia](#) [Radio Wrocław](#) — to sophisticated hi-fi stereo systems in the 1970s and 80s. The factory produced the **Meluzyna WST-101**, Poland's first transistor hi-fi set (1972), [Oldradio](#) [Audio-room](#) and the **Elizabeth** line developed in collaboration with Tokyo-Sanyo Electric of Japan (1973). [Wikipedia](#) It also manufactured approximately **600,000 television sets** between 1959 and 1969, [Oldradio](#) military radio stations for Polish Army tanks, car radios, VHS video recorders, and satellite TV tuners. [Wikipedia](#)

From the 1960s through the 1980s, Diora operated under the **Unitra** umbrella [Wikipedia](#) — Poland's state electronics conglomerate encompassing 28 companies and roughly 100,000 workers. [Grokikipedia](#) Its products were exported to Western Europe and Canada, and it maintained international collaborations with [Wedrownemisie](#) AGA-Baltic (Sweden), [Letheko](#) Thomson-Brandt (France), [Wikipedia](#) and [Mmdz](#) Siemens (Germany). [Wedrownemisie](#) The factory ran its own technical school, the "Radiobudowa," training future electronics cadres from [Mmdz](#) 1946 onward. [Mmdz](#) [Oldradio](#)

The connection to Bielawa is intimate but indirect. Diora was not in Bielawa itself but in neighboring Dzierżoniów — the two towns sit in the same county, [RFN](#) connected by a railway line since 1891, [Wikipedia](#) separated by a short bus ride. [Rome2rio](#) With over 6,600 employees at peak and Dzierżoniów being a small city, many workers commuted from surrounding towns including Bielawa. Pete's grandfather Józef working at Diora would have placed him in the heart of the region's electronics culture — surrounded by radio circuits, amplifiers, and the technical knowledge to understand and appreciate a home computer.

Diora did not survive the post-Communist transition. Privatized in 1989 as Diora S.A., the factory hemorrhaged workers through the 1990s (Oldradio) — from thousands to roughly **200 employees by 1999**. (Wedrownemisie) It was struck from the business register in **2006**, and its buildings were demolished. (Wikipedia) A **Kaufland supermarket** now stands on the site. The street was renamed "Diorowska" in commemoration, (Wikipedia) and since 2018, Dzierżoniów has maintained a "Trakt Diory" (Diora Trail) — a tourist path with ten brass 1:1 scale radio replicas placed around the town center. (Dzierzoniow) (Naszemiasto)

Growing up in Bielawa: a textile town in the Owl Mountains

Bielawa (German: Langenbielau) (Wikipedia) sits in **Lower Silesian Voivodeship**, southwestern Poland, (Wikipedia) nestled at the foot of the **Owl Mountains** (Góry Sowie) (Omicsgroup) (En Academic) about **56 km south of Wrocław**. (Entfernungsrechner) (Wikipedia) Its population was approximately **32,000** in the era Pete grew up there, declining to around **28,300** by 2023. (Wikipedia) The town's identity was forged by textiles — the **Christian Dierig** weaving company, founded in 1805, (Wikipedia) grew to operate 21 textile mills employing over 15,000 people across Lower Silesia by World War II. (Palacbielawa) Bielawa was also the site of the famous **1844 Silesian Weavers' Uprising**, (Wikipedia) one of the first major labor revolts in the German-speaking world, immortalized by Heinrich Heine's poem and Gerhart Hauptmann's play *The Weavers*. (Wikipedia)

The town's modern Polish identity began in 1945 when the German population was expelled and replaced by Polish settlers from diverse origins — former eastern territories (including Kołomyja in present-day Ukraine), central Poland, and Poles returning from France, Germany, and Romania. (Academia.edu) Greek refugees from the Greek Civil War also settled there. (Wikipedia) This created a community built "on raw roots," relatively young and with mixed regional identities. (Academia.edu) Families literally moved into German houses and used German furniture; the inherited material culture was everywhere, though its origins were officially suppressed.

(Cambridge Core)

Growing up in Bielawa in the 1980s meant living in a small industrial town dominated by state-run textile factories, in the shadow of Communist-era austerity. Basic consumer goods were scarce — chocolate bars were a luxury, Coca-Cola was available only in Warsaw and Katowice. (Myapplemagazine) The economy was failing. After 1989, the transition brought further devastation: Poland's electronics sector production fell **25% in 1990 alone**, (Grokipedia) and state-run factories across towns like Bielawa and Dzierżoniów faced collapse. Bielawa's textile industry ultimately ceased operations in **2008**. (Academia.edu)

The town today brands itself as a "Model Ecological Town," with its **Neo-Gothic Church of the Assumption** (sporting Poland's **third-tallest church tower** at 101 meters), (HelloTravel) (Wikipedia) a lakeside recreation area, and trails into the Owl Mountains Landscape Park. (Omicsgroup) (Davidstravelguides)

A ZX Spectrum in Communist Poland: magic in a rubber keyboard

The Sinclair ZX Spectrum [\(Wikipedia\)](#) 48K, released in Britain on **April 23, 1982**, [\(Wikipedia\)](#) was a deliberately cheap home computer: a **Zilog Z80A** processor running at 3.5 MHz, [\(Wikipedia\)](#) 48 KB of RAM, [\(Riot\)](#) 15 colors, single-channel beeper sound, [\(Wikipedia\)](#) and the famous rubber "dead flesh" keyboard. It stored programs on ordinary audio cassette tapes and connected to a standard television. What made it revolutionary was accessibility — it cost just £175 at launch, and the moment you turned it on, you were at a **BASIC programming prompt**, ready to code. [\(Medium\)](#)

Getting one into Poland was another matter entirely. The COCOM embargo restricted Western technology exports to the Eastern Bloc, [\(Grokikipedia\)](#) [\(Myapplemagazine\)](#) though Poland received somewhat milder treatment "in hopes of driving a wedge between the Soviet Union and its allies." [\(American Foreign Relations\)](#) Computers entered Poland through a patchwork of channels: individuals smuggling them from the West, [\(Myapplemagazine\)](#) **Pewex** hard-currency shops (state-run stores selling Western goods for dollars), and eventually the **Timex Computer 2048** — a Spectrum-compatible clone manufactured by Timex of Portugal and sold in Poland from **1986** through the Centralna Składnica Harcerska (Central Scouting Store), with 5,000 units initially available at **PLZ 106,000** — equivalent to roughly **4.4 average monthly salaries**. An original ZX Spectrum cost PLZ 70,000–80,000 on the grey market, still about **3 monthly salaries**. [\(Wikipedia\)](#) The Polish-made **Elwro 800 Junior**, a Spectrum clone designed for schools [\(Wikipedia\)](#) (approximately 13,700 units produced), [\(Sinclaircollection\)](#) was famously housed in the case of a toy electronic keyboard and came with a built-in classroom networking system. [\(Infogalactic\)](#) [\(Wikipedia\)](#)

Having a home computer in a small Polish town in the late 1980s was profoundly extraordinary. Many readers of *Bajtek* — Poland's iconic computer magazine, launched in 1985, whose first 200,000 copies sold instantly [\(CPCWiki\)](#) — didn't even own a computer and read about computing vicariously. The magazine functioned as a "virtual computer" for an entire generation. [\(Pismowidok\)](#) Software distribution was ingeniously improvised: cassette tape trading at bazaars, programs typed in from magazine listings, and remarkably, a radio show called *Radio Komputer* that **broadcast computer programs over the airwaves** — listeners with a radio-cassette recorder could capture the audio signal and load it onto their Spectrum. At least one Polish pop band released a vinyl record whose last track was the audio encoding of a ZX Spectrum program. [\(Myapplemagazine\)](#)

For Józef, a Diora electronics worker with deep familiarity with circuits and radio technology, acquiring a ZX Spectrum was both a natural extension of his professional world and an act of extraordinary investment. In a village where few if any other households had a computer, the machine would have been an object of fascination. The experience of learning BASIC on the Spectrum was uniquely tactile: each of the 40 rubber keys was mapped to multiple BASIC keywords (pressing "P" at the start of a line typed the whole word [\(PRINT\)](#)), [\(Wikipedia\)](#) so a child could quickly produce results. [\(Riot\)](#) A five-year-old could make text cascade down the screen with [\(10 PRINT "HELLO"\)](#) / [\(20 GOTO 10\)](#) / [\(RUN\)](#). By age seven or eight, [\(FOR...NEXT\)](#) loops, [\(IF...THEN\)](#) logic, [\(PLOT\)](#) and [\(DRAW\)](#) graphics commands, and [\(BEEP\)](#) sound effects were all within reach. Loading programs meant enduring minutes of screeching cassette audio [\(Codeguppy\)](#) and the dreaded "tape loading error" [\(Myapplemagazine\)](#) — teaching patience alongside programming. [\(Myapplemagazine\)](#)

From Commodore to Amiga to PC: Poland's wild computing progression

The path Pete describes — ZX Spectrum → Commodore → Amiga → early PCs → consoles — mirrors the broader trajectory of an entire Polish generation, running roughly **2–5 years behind Western Europe** at every stage due to economic constraints and the Iron Curtain.

The Commodore 64 arrived in Poland in the mid-1980s through smuggling and Pewex shops, with no official distribution. Its golden age in Poland ran approximately **1986–1994**. Software distribution was entirely pirate-driven — the legendary distributor Waldemar Czajkowski mass-produced pirated cassette tapes in runs of **12,000+ copies**, driving them around Poland in vans, earning enough to buy an apartment and his first car.

[River's Edge](#) There were no copyright laws prohibiting any of this until **1994**.

The Amiga became Poland's aspirational computing platform in the early 1990s. The Amiga 500 started reaching Poland around 1988, imported by small-scale traders with contacts in West Germany. [widerscreen](#) By the end of 1994, an estimated **100,000–120,000 Amiga computers** had been sold in Poland. [widerscreen](#)

[Docslib](#) The machine spawned a thriving **demoscene** — Poland had one of Eastern Europe's most active, hosting events like "Intel Outside" (a cheeky inversion of Intel's marketing slogan) [widerscreen](#) and producing over **250 local disk magazines** between 1991 and 1995. [widerscreen](#)

The **gielda komputerowa** (computer bazaar) was the beating heart of this culture. The most famous, on ulica Grzybowska in Warsaw, launched in spring 1986 [SaveTheFloppy](#) under *Bajtek* magazine's patronage. [Polygamia](#) These weekend bazaars were simultaneously marketplace, social club, and technology expo — described by participants as "our university, our research center, and our meeting place." [SilverPC Blog](#) Pirate software sellers were community pillars who made computing accessible. [Gadzetomania](#) Two of the most consequential figures in Polish gaming history emerged from this scene: **Marcin Iwiński**, a software cracker, and **Michał Kiciński**, a pirate software vendor from the Grzybowska fair — they co-founded **CD Projekt**, the studio behind *The Witcher* and *Cyberpunk 2077*. [VOGONS](#)

The PC transition came around 1994–1995, accelerated by Commodore's bankruptcy and the CD-ROM revolution. The magazine **CD-Action**, launched in April 1996, came bundled with CDs containing game demos and full versions [Wikipedia](#) [TheReaderWiki](#) — a powerful draw in a market where buying games was expensive. Poland's unique console experience was the **Pegasus** — an unauthorized Taiwanese-made Famiclone (NES clone) sold from 1991 [Wikipedia](#) that became so ubiquitous the word "Pegasus" entered Polish as a generic term for any game console, [Retro Junk](#) much as "Nintendo" did in the United States. Poland essentially **skipped the major console generations** (NES, SNES, Mega Drive were never officially sold there), which is why it remains famously a "PC gaming country" to this day. [Game Informer](#)

World of Warcraft: the game that consumed a generation

When Pete names World of Warcraft as the game he spent the most hours on, he's naming **the most successful MMORPG ever made**. Launched on **November 23, 2004** by Blizzard Entertainment, (EBSCO) WoW peaked at **12 million subscribers** (Activision Blizzard) in October 2010 and accumulated **100 million registered accounts** by 2014, generating over **\$9.23 billion** in revenue by 2017. (Wikipedia) At its zenith, it commanded an estimated **62% of the entire MMORPG subscription market**. (Wikipedia) The game's design — an expansive 3D world combined with guild-based social mechanics, coordinated raid content requiring dozens of players, and a compelling progression treadmill — created a famously time-consuming experience. Players collectively logged over **a billion hours** of gameplay (Complex) by 2014. The game has received ten major expansions across 20+ years (Lagofast) and remains active, (Playstorypress) with an eleventh expansion (*Midnight*) scheduled for early 2026.

Chompd: Pete's first app, built without writing code

Chompd (at getchompd.com) is Pete Michaliszyn's first software application, and it appears to be **extremely new — likely pre-launch or in earliest release** as of February 2026. A Google Drive folder for the project was created on **February 15, 2026**, just days before this research. The app has no public web footprint yet: no search engine indexing, no app store listings, no press coverage, and no social media mentions were found. The name and domain strongly suggest a **food or restaurant-related application**, though specific features could not be confirmed from public sources.

What makes Chompd biographically significant is not (yet) the app itself but the method of its creation. Pete — whose professional career has been as a **World Cup downhill mountain bike mechanic**, serving as the personal "wrencher" for elite racer **Gee Atherton** on the Trek Factory Racing team for at least seven years — has **no formal programming background**. He founded **Polished Racing Ltd** in 2015 (now dissolved), (service) which served as a sponsor of the Atherton Racing team alongside brands like Red Bull, Trek, and Shimano. He was also involved in the early development of **Atherton Bikes**, revolutionary downhill frames using 3D titanium-printed lugs, where he completed prototype builds and oversaw testing. (IR|evolution MTB)

Evidence from his Google Drive indicates he is actively working with **Anthropic's Claude AI models** — a document titled "OpenClaw Token Optimization Guide" (created February 6, 2026) discusses AI API cost optimization including token reduction and model routing between Claude Haiku and Sonnet. This confirms his use of AI coding tools consistent with the "vibe coding" methodology he describes.

Vibe coding: when English becomes a programming language

Vibe coding — the practice of building software by describing what you want in plain language and letting AI generate the code — was coined by **Andrej Karpathy**, (Graphite) co-founder of OpenAI and former AI director at Tesla, (Medium) in a post on X on **February 6, 2025**. His description was characteristically casual: *"You fully give in to the vibes, embrace exponentials, and forget that the code even exists... (Time) I 'Accept All' always, I don't read the diffs anymore... (X) it's not really coding — I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works."* (Medium) (Klover)

The term exploded. **Collins English Dictionary** named "vibe coding" its **Word of the Year for 2025**.

[Wikipedia](#) [CNN](#) Y Combinator reported that **25% of startups in its Winter 2025 batch** had codebases that were **95% AI-generated**. [RapidNative](#) [Wikipedia](#) Google CEO Sundar Pichai admitted to vibe coding his own website for fun. [Entrepreneur](#) [Businesschief](#) New York Times journalist Kevin Roose built **LunchBox Buddy** — an app that analyzes fridge photos and suggests children's lunch ideas [Instamobile](#) — in roughly 10 minutes. [Redwerk](#) Non-programmers across professions began building functional software: an HR professional on maternity leave created a family meal planner, [The Wave](#) [Redwerk](#) a marketing professional built a journaling app for stressed parents, [Redwerk](#) a Harvard neuroscience student created an ancient text translator. [Entrepreneur](#)

The key tools enabling vibe coding include **Cursor** (an AI-powered code editor that became the fastest app to reach \$100M revenue, doing so in just two months), [Substack](#) **Claude** (Anthropic's LLM, specifically mentioned in Karpathy's original tweet), **Replit**, **Lovable**, and **GitHub Copilot**. The critical distinction, as articulated by developer Simon Willison: if you review, test, and understand every line the AI writes, that's just using AI as a typing assistant. Vibe coding specifically means **accepting code you haven't fully read or understood** — trusting the output based on whether the result works. [Wikipedia](#)

Criticism has been substantial. A December 2025 CodeRabbit analysis of 470 open-source GitHub pull requests found AI co-authored code contained **2.74x more security vulnerabilities** and **75% more logic errors** than human-written code. [Wikipedia](#) A survey of 18 CTOs found 16 reporting production disasters caused by AI-generated code. [Medium](#) By February 2026, Karpathy himself had moved on, [Startup News](#) proposing "agentic engineering" as a more mature successor concept — emphasizing that orchestrating AI agents to write code is itself a skill requiring expertise and oversight. [The New Stack](#)

For Pete's story, vibe coding completes a striking circle. A child who learned [PRINT "HELLO"](#) on a rubber-keyed ZX Spectrum in a Polish village — where the computer itself was an act of extraordinary acquisition — now builds software by speaking to an AI in plain English, never touching a programming language directly. The grandfather who worked at a radio factory taught the five-year-old the foundational insight: a machine does what you tell it. Forty years later, the language of instruction has changed from BASIC to English, but the principle holds. [Medium](#)